

3. Write a Java class called Tax with methods for calculating Income Tax. Have this class as a servant and create a server program and register in the rmiregistry. Write a client program to invoke these remote methods of the servant and do the calculations. Accept inputs interactively.

Remote interface tax

```
interface tax extends java.rmi.Remote
{
    public double cal(double amt) throws java.rmi.RemoteException;
}
```

Tax implementation

```
class taximpl extends java.rmi.server.UnicastRemoteObject implements
tax
```

```
{
    taximpl() throws java.rmi.RemoteException
    {
        super();
    }
    public double cal(double amt)
    {
        if(amt<=300000)
            return(0.0);
        else if(amt<=600000)
            return(amt*0.05);
        else if(amt<=900000)
            return(amt*0.1);
        else if(amt<=1200000)
            return(amt*0.15);
        else if(amt<=1500000)
            return(amt*0.2);
        else
            return(amt*0.3);
    }
}
```

Server Program

```
import java.rmi.Naming;
class taxserver
{
    taxserver()
    {
```

```
        try
        {
            tax tx=new taximpl();
            Naming.rebind("rmi://localhost:1099/taxservice",tx);
        }
        catch(Exception e)
        {
            System.out.println(e.getMessage());
        }
        System.out.println("Tax server is running");
    }
    public static void main(String args[])
    {
        new taxserver();
    }
}
```

Client Program

```
import java.io.*;
import java.rmi.Naming;
class taxclient
{
    public static void main(String args[])
    {
        DataInputStream in=new DataInputStream(System.in);
        double anual;
        try
        {
            tax t=(tax)Naming.lookup("rmi://localhost/taxservice");
            System.out.println("Enter anual income");
            anual=Double.parseDouble(in.readLine());
            System.out.println("Tax is "+t.cal(anual));
        }
        catch(Exception e)
        {
            System.out.println(e.getMessage());
        }
    }
}
```

OUTPUT:

```
D:\jdk1.6.0\bin>javac tax.java
```

```
D:\jdk1.6.0\bin>javac taximpl.java
```

```
D:\jdk1.6.0\bin>javac taxserver.java
```

```
D:\jdk1.6.0\bin>javac taxclient.java
```

Note: taxclient.java uses or overrides a deprecated API.

Note: Recompile with -Xlint:deprecation for details.

```
D:\jdk1.6.0\bin>start rmiregistry
```

```
D:\jdk1.6.0\bin>rmic taximpl
```

```
D:\jdk1.6.0\bin>java taxserver
```

Tax server is running

```
D:\jdk1.6.0\bin>java taxclient
```

Enter anual income

300000

Tax is 0.0

```
D:\jdk1.6.0\bin>java taxclient
```

Enter anual income

500000

Tax is 25000.0

```
D:\jdk1.6.0\bin>java taxclient
```

Enter anual income

700000

Tax is 70000.0

```
D:\jdk1.6.0\bin>java taxclient
```

Enter anual income

900000

Tax is 90000.0

```
D:\jdk1.6.0\bin>java taxclient
```

Enter anual income

1200000

Tax is 180000.0

```
D:\jdk1.6.0\bin>java taxclient
```

Enter anual income

1600000

Tax is 48000

4. Write a Java class called SimpleInterest with methods for calculating simple interest. Have this class as a servant and create a server program and register in the rmiregistry. Write a client program to invoke these remote methods of the servant and do the calculations. Accept inputs at command prompt.

Remote interface simple

```
interface simple extends java.rmi.Remote
{
    public double sical(double p,double t,double r) throws
    java.rmi.RemoteException;
}
```

Simple implementation

```
class simpleimpl extends java.rmi.server.UnicastRemoteObject implements
simple
{
    simpleimpl() throws java.rmi.RemoteException
    {
        super();
    }
    public double sical(double p,double t,double r)
    {
        return(p*t*r/100);
    }
}
```

Server Program

```
import java.rmi.Naming;
class siserver
{
    siserver()
    {
        try
        {
            simple si=new simpleimpl();
            Naming.rebind("rmi://localhost:1099/siservice",si);
        }
        catch(Exception e)
        {
            System.out.println(e.getMessage());
        }
    }
}
```

```
        }
        System.out.println("Simple interest server is running");
    }
    public static void main(String args[])
    {
        new siserver();
    }
}
```

Client Program

```
import java.io.*;
import java.rmi.Naming;
class siclient
{
    public static void main(String args[])
    {
        double p,t,r;
        try
        {
            if(args.length==3)
            {
                simple si=(simple)Naming.lookup("rmi://localhost/siservice");
                p=Double.parseDouble(args[0]);
                t=Double.parseDouble(args[1]);
                r=Double.parseDouble(args[2]);
                System.out.println("The simple interest is:"+si.sical(p,t,r));
            }
            else
            {
                System.out.println("insufficient arguments");
            }
        }
        catch(Exception e)
        {
            System.out.println(e.getMessage());
        }
    }
}
```

OUTPUT:

```
D:\jdk1.6.0\bin>javac simple.java
D:\jdk1.6.0\bin>javac simpleimpl.java
D:\jdk1.6.0\bin>javac siserver.java
D:\jdk1.6.0\bin>javac siclient.java
D:\jdk1.6.0\bin>start rmiregistry
D:\jdk1.6.0\bin>rmic simpleimpl
D:\jdk1.6.0\bin>java siserver
Simple interest server is running
D:\jdk1.6.0\bin>java siclient
insufficient arguments

D:\jdk1.6.0\bin>java siclient 3000 5 7
The simple interest is:1050.0
```

8. Implement a java application based on the MVC design pattern. Input student Rolnlo, name ,marks in three subject calculate result and grade and display the result in neat format

Student Class

```
package stud;
```

```
public class Student
{
    private String sname;
    private int rollno,mark1,mark2,mark3;
    public String getSname()
    {
        return sname;
    }
    public void setSname(String name)
    {
        this.sname = name;
    }
    public int getRollno()
    {
        return rollno;
    }
    public void setRollno(int rollno)
    {
        this.rollno = rollno;
    }
    public int getMark1()
    {
        return mark1;
    }
    public void setMark1(int mark1)
    {
        this.mark1 = mark1;
    }
    public int getMark2()
    {
        return mark2;
    }
    public void setMark2(int mark2)
    {
        this.mark2 = mark2;
    }
    public int getMark3()
    {
        return mark3;
    }
    public void setMark3(int mark3)
    {
        this.mark3 = mark3;
    }
}
```

```
    }  
}
```

StudentView code

```
package stud;  
  
public class StudentView  
{  
    public void display(int rollno,String name,int mark1,int mark2,int  
mark3)  
    {  
        int total;  
        double per;  
        String grade;  
        total=mark1+mark2+mark3;  
        per=total/3;  
        if(per>90)  
            grade="A";  
        else if(per>80)  
            grade="B";  
        else if(per>70)  
            grade="C";  
        else if(per>60)  
            grade="D";  
        else  
            grade="E";  
        System.out.println("*****Student  
Detail*****");  
        System.out.println("Student Name:"+name);  
        System.out.println("Student Roll Number:"+rollno);  
        System.out.println("Mark1:"+mark1);  
        System.out.println("Mark2:"+mark2);  
        System.out.println("Mark3:"+mark3);  
        System.out.println("Total Marks:"+total);  
        System.out.println("Average MArks:"+per);  
        System.out.println("Student Grade:"+grade);  
  
    }  
}
```

Student Control Code


```
package stud;
```

```
public class StudentControl
{
    private Student model;
    private StudentView view;
    public StudentControl(Student model, StudentView view)
    {
        super();
        this.model = model;
        this.view = view;
    }
    public void setStudRoll(introllno)
    {
        model.setRollno(rollno);
    }
    public intgetStudRoll()
    {
        return model.getRollno();
    }
    public void setStudName(String name)
    {
        model.setSname(name);
    }
    public String getStudName()
    {
        return model.getSname();
    }
    public void setStudMark1(int mark1)
    {
        model.setMark1(mark1);
    }
    public int getStudMark1()
    {
        return model.getMark1();
    }
    public void setStudMark2(int mark2)
    {
        model.setMark2(mark2);
    }
    public int getStudMark2()
    {
        return model.getMark2();
    }
}
```

```
    }  
    public void setStudMark3(int mark3)  
    {  
        model.setMark3(mark3);  
    }  
    public int getStudMark3()  
    {  
        return model.getMark3();  
    }  
    public void updateView()  
    {  
view.display( model.getRollno(),model.getSname(), model.getMark1(),  
model.getMark2(), model.getMark3());  
    }  
}
```

Student Application Program

```
package stud;
```

```
import java.io.DataInputStream;
```

```
import java.io.IOException;
```

```
public class Demo {
```

```
    public static void main(String[] args) throws IOException {  
        // TODO Auto-generated method stub  
        Student model=retrieveStudent();  
        StudentView view=new StudentView();  
        StudentControl control=new StudentControl(model,view);  
        control.updateView();  
    }
```

```
    private static Student retrieveStudent()throws IOException  
    {
```

```
        String sname;  
        Int rollno,mark1,mark2,mark3;  
        DataInputStream in=new DataInputStream(System.in);  
        System.out.println("Enter Student Name");  
        sname=in.readLine();  
        System.out.println("Enter Roll Number");  
        rollno=Integer.parseInt(in.readLine());  
        System.out.println("Enter Mark1");  
        mark1=Integer.parseInt(in.readLine());
```

```
        System.out.println("Enter Mark2");
        mark2=Integer.parseInt(in.readLine());
        System.out.println("Enter Mark3");
        mark3=Integer.parseInt(in.readLine());
        Student stud=new Student();
        stud.setRollno(rollno);
        stud.setName(sname);
        stud.setMark1(mark1);
        stud.setMark2(mark2);
        stud.setMark3(mark3);
        return stud;
    }
}
```

OUTPUT:

```
Enter Student Name
vamshi
Enter Roll Number
101
Enter Mark1
95
Enter Mark2
95
Enter Mark3
97
*****Student
Detail*****
Student Name:vamshi
Student Roll Number:101
Mark1:95
Mark2:95
Mark3:97
Total Marks:287
Average MARKs:95.0
Student Grade:A
```

```
Enter Student Name
kavana
Enter Roll Number
103
Enter Mark1
70
```

```
Enter Student Name
varun
Enter Roll Number
102
Enter Mark1
85
Enter Mark2
90
Enter Mark3
85
*****Student
Detail*****
Student Name:varun
Student Roll Number:102
Mark1:85
Mark2:90
Mark3:85
Total Marks:260
Average MARKs:86.0
Student Grade:B
```

```
Enter Mark2
75
Enter Mark3
76
*****Student
Detail*****
```

Student Name:kavana
Student Roll Number:103
Mark1:70
Mark2:75
Mark3:76
Total Marks:221
Average MARKs:73.0
Student Grade:C

Student Roll Number:105
Mark1:50
Mark2:55
Mark3:57
Total Marks:162
Average MARKs:54.0
Student Grade:

Enter Student Name
varsha
Enter Roll Number
104
Enter Mark1
65
Enter Mark2
66
Enter Mark3
67
*****Student
Detail*****
Student Name:varsha
Student Roll Number:104
Mark1:65
Mark2:66
Mark3:67
Total Marks:198
Average MARKs:66.0
Student Grade:D

Enter Student Name
radhika
Enter Roll Number
105
Enter Mark1
50
Enter Mark2
55
Enter Mark3
57
*****Student
Detail*****
Student Name:radhika

III BCA

PART-B

ROLL NO:

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